

CoMIC: A Collaborative Mobile Immersive Computing Infrastructure for Conducting Multi-User XR Research

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ABSTRACT

While existing research on immersive technologies such as virtual, augmented, and mixed reality has been shifting from improving the quality of experience of a single user to multi-user scenarios, which are naturally required by the burgeoning Metaverse, there is little work on building the underlying infrastructure to support multi-user immersive applications, especially for geo-distributed users. In this position paper, we propose a research infrastructure, dubbed CoMIC, to fill this critical gap, by offering a collaborative, visual-first, and hologram-based computing space and a QoE-driven, multi-site, and immersive communication framework. With its rudimentary prototype, we demonstrate the effectiveness of CoMIC through a case study of multi-user volumetric video streaming over mmWave. We conclude this paper with a discussion of numerous unique opportunities that can be enabled by CoMIC.

INTRODUCTION

The last few years have witnessed the tremendous development of immersive technologies such as virtual, augmented, and mixed reality (VR, AR, and MR) [3], [11], [14], which are often referred to as a universal term, extended reality (XR). Along with XR, the burgeoning Metaverse [9] has appeared in almost every list of tech trends in 2022. Although there is no consensus on the definition, a narrow depiction of the Metaverse is a ubiquitous virtual world that facilitates social interaction among users, by connecting multiple 3D virtual environments via the Internet (i.e., a network of virtual worlds).

While XR is deemed as the killer application of 5G and beyond, existing work mainly focused on improving the quality of experience (QoE) for single-user scenarios [3], [11], [14]. For example, in addition to the common QoE metrics of traditional video streaming such as the startup delay, video encoding bitrate, resolution of video frames, and rebuffering time, recent studies [5], [7], [11] demonstrate that the QoE of immersive technologies is also influenced by other factors including content visibility, viewing distance, lighting conditions, visual acuity, motion-to-photon latency, motion sickness, etc. However, there is limited research on exploring the *underlying infrastructure*

to support emerging immersive applications, especially multi-user collaborations and interactions involving participants from geo-distributed locations, which is naturally required by the Metaverse and various use cases such as education, training, entertainment, healthcare, retail, etc.

The XR research community faces three major challenges regarding (1) software framework, (2) testbeds, and (3) tools, resources, and datasets for conducting multi-user XR research, which is more difficult than single-user XR. For example, as demonstrated in Coterie [8] (a multi-user VR system), simply extending existing solutions for single-user VR to multi-user scenarios will lead to the linear increase of both transmission latency and network throughput when the number of users grows. Moreover, to smooth the interaction of multiple users with the same virtual content, multi-user AR should consider more QoE metrics than the single-user counterpart, such as content synchronization accuracy, interaction delay, the difference and correlation of multi-user frame rate and content resolution, etc.

- *Software Framework.* There is no software framework with commonly used building blocks for designing and implementing multi-site, multi-user XR applications. Given its interdisciplinary nature, developing such a framework requires expertise from different research disciplines. For example, for the synchronization of instantaneous interaction among users, especially across different sites, there should be an efficient middleware at the network edge that can construct in real time spatial maps with computer vision (CV) techniques [10]. To improve the scalability of multi-user XR, high-performance computer graphics algorithms should be deployed on edge servers to remotely render immersive content [11]. Although existing software frameworks such as Google ARCore and Apple ARKit have started to enable the sharing of experiences among multiple users, they mainly investigate a single aspect of multi-user XR (i.e., synchronization among users^{1,2}) in their initial efforts.
- *Testbed.* There is no remotely accessible and reconfigurable testbed for testing, evaluating, and benchmarking multi-user XR technologies and applications. For example, the recently proposed ILLIXR³ is a testbed to

¹ <https://developers.google.com/ar/develop/cloud-anchors> (accessed on 10-August-2022).

² https://developer.apple.com/documentation/arkit/creating_a_multiuser_ar_experience (accessed on 10-August-2022).

³ <https://illixr.github.io> (accessed on 10-August-2022).

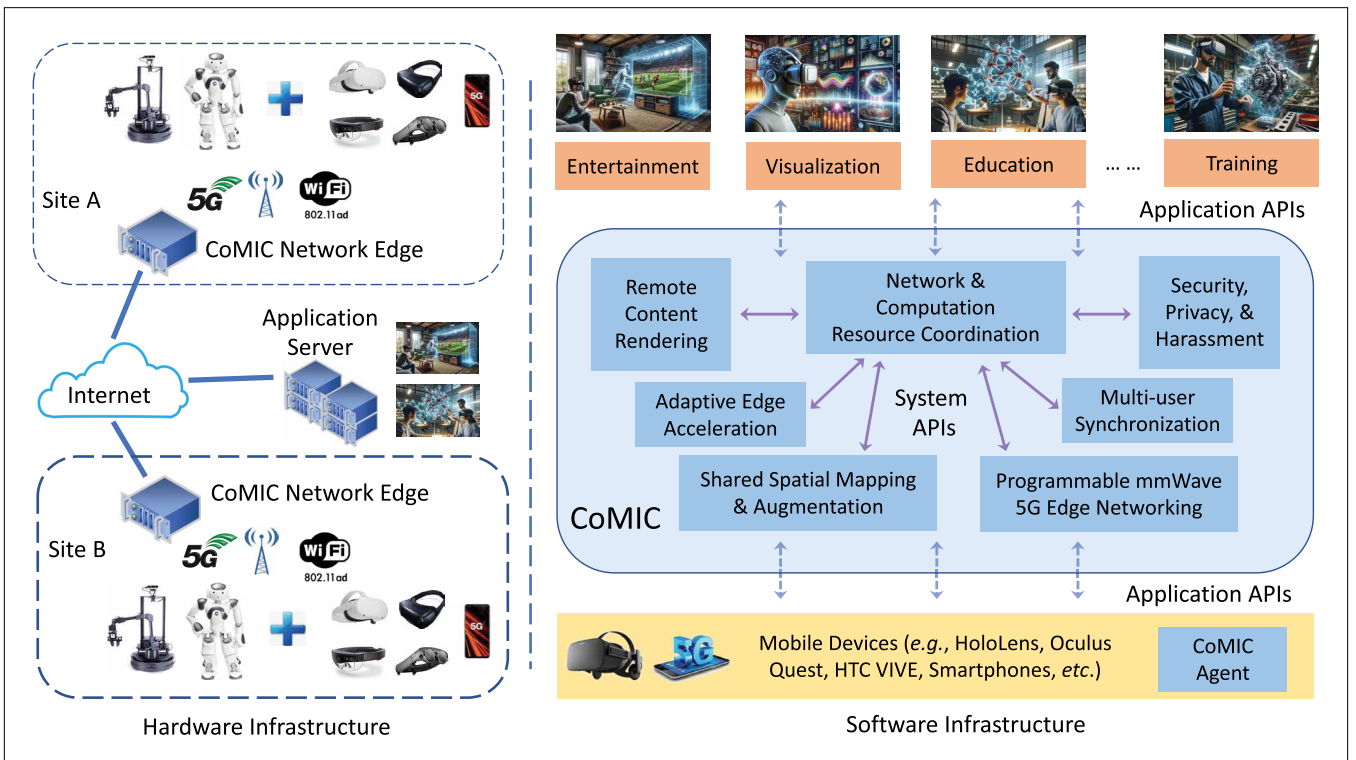


FIGURE 1. The proposed CoMIC system infrastructure. The interactions among the six key components are coordinated by the central scheduler (§III-A).

facilitate the development of single-user XR applications. Due to the data-intensive nature of immersive computing [6], the testbed should be equipped with high-throughput networks (e.g., programmable mmWave 5G) to facilitate the real-time delivery of immersive content. Also, to satisfy the latency requirements of multi-user synchronization and remote content rendering in XR, the testbed should be located at the network edge that is close to end-users. Moreover, to reduce the burden of recruiting participants for evaluating human-centric XR, the testbed should be capable of automating experiments for remote users, without human intervention, in a reproducible fashion.

- **Tools, Resources, and Datasets.** There is a lack of commonly used tools, resources, and datasets, which severely hinders the development and evaluation of multi-user XR systems. For example, there are no widely acceptable tools to measure the synchronization accuracy of multiple users with a common coordinate system in the shared 3D space [10], an issue that does not exist for single-user XR. Also, the datasets for evaluating 6DoF (six degrees of freedom) motion prediction are still limited to single-user scenarios [7], although leveraging the potential correlation between the trajectories of multiple users when watching the same volumetric video at the same time may improve the accuracy of motion prediction. Moreover, there are no well-established QoE models to quantify multi-user XR experiences, as modeling QoE for multi-user XR is still an open problem. Although existing

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work typically targets the QoE of single-user XR applications [3], [11], [14] and most of the QoE metrics for single-user XR are still applicable to the multi-user counterpart, the interactions among users bring more challenges and opportunities for modeling and improving the QoE for multi-user XR, because the experience of one user may be affected by and correlated with that of others.

While there have been some recent endeavors on boosting the performance of multi-user VR, AR, and volumetric video streaming, they usually focus on a single point in the design space, such as high-bandwidth demand [8], [13], multi-user synchronization [10], and scalability challenges [15]. It is still unclear how to build a holistic system that can effectively integrate individual optimizations, which is non-trivial due to the potential conflicts between the point solutions. In this paper, we propose, to the best of our knowledge, the first-of-its-kind research infrastructure for Collaborative Mobile Immersive Computing (CoMIC), with the overarching goal of advancing multi-user XR research. We emphasize that the goal of this position paper is not to directly address the research challenges of multi-user XR. Instead, we have identified the lack of the underlying infrastructure for supporting multi-user XR research and, to this end, proposed the CoMIC infrastructure.

Building Blocks	System APIs	Application APIs
Programmable mmWave 5G Edge Networking	mmWave sensing for estimating 6DoF pose of XR headsets	mmWave MAC/PHY reconfiguration & beam adaptation
Adaptive Edge Acceleration	GPU scheduler for computer vision & deep learning tasks	GPU-accelerated computer vision & deep learning modules
Multi-user Synchronization	Relative & absolute positions of users	Synchronized positions & trajectories, distance between objects
Remote Content Rendering	Multi-user 6DoF motion prediction	Scalable neural-enhanced multiview synthesis
Shared Spatial Mapping and Augmentation	Spatial map and semantic information of environment	Common free space among users, placement of shared virtual content
Security, Privacy, and Harassment	Privacy-aware user activity tracking, S&P policies	Privacy-preserving authentication, harassment detection & prevention

TABLE 1. Major system and application APIs for CoMIC components.

OVERVIEW OF CoMIC

CoMIC addresses the above crucial challenges and lays the technical foundation for realizing the grand vision of the Metaverse. It strives to empower a *collaborative, visual-first, and hologram-based* computing space and design a *QoE-driven, multi-site, and immersive* communication framework. CoMIC features several key capabilities for conducting multi-user XR research. As shown in Fig. 1, the CoMIC software runs on the edge server that is deployed for each site or shared by closeby sites.

The CoMIC software consists of the following six key building blocks for multi-user XR.

1. CoMIC aims at developing a programmable mmWave 5G edge networking framework that enables agile, cross-layer, and resilient interfaces for high-speed, low-latency, and multi-user immersive content delivery through real-time PHY/MAC reconfiguration.
2. CoMIC will adaptively offload computation-intensive tasks of XR (e.g., 3D object detection and semantic segmentation) to GPU-accelerated edge servers for reducing end-to-end latency and improving QoE [14].
3. Given its multi-user, multi-site nature, CoMIC will synchronize the interactions of geographically distributed users with the displayed holographic content and create a common perspective within a collaborative session [15].
4. To alleviate the computation overhead of rendering high-quality holographic content on mobile devices [1], CoMIC seeks to pre-render the content for multiple users in a scalable fashion on the edge server based on the predicted 6DoF motion of users.
5. By sharing spatial maps, CoMIC will ensure smooth collaborations among users as virtual objects are augmented at each user's site in a collision-free manner by taking into account the local scene geometry and semantics.
6. CoMIC strives to protect users from security threats and preserve their privacy in the shared environment. It will exercise appropriate privacy-preserving authentications to protect the digital assets of users and shield them from harassment.

Besides the six components on the edge server, the on-device CoMIC agent is mainly in charge of collecting runtime data such as the 6DoF motion of users and configuring the wireless network interface if needed.

The CoMIC testbed initially consists of two sites. Each site will be equipped with multiple XR headsets. We will use robots to facilitate access for remote users. These sites will be equipped with programmable mmWave 5G networks for experiments with data-intensive immersive content delivery in multi-user XR. CoMIC will offer a comprehensive set of tools such as remote controllers of XR devices, robots, and mmWave 5G networks and various datasets (e.g., multi-user 6DoF motion traces).

THE CoMIC RESEARCH INFRASTRUCTURE

In this section, we will present the CoMIC software framework and its tools and datasets in detail.

SOFTWARE FRAMEWORK

We list the major system and application APIs of each CoMIC building block in Table 1. Note that some of the system APIs (e.g., 6DoF motion prediction) can also be utilized by multi-user XR applications. The coordination of interactions among the CoMIC components is taken care of by a central scheduler. As a holistic software framework, we give a few examples of the interactions of these building blocks. The multi-user 6DoF motion prediction API offered by the "Remote Content Rendering" can be utilized by the "Programmable mmWave 5G Edge" for blockage mitigation. The maps generated by the "Shared Spatial Mapping and Augmentation" will be helpful to determine ambient reflectors for the "Programmable mmWave 5G Edge". The relative and absolute positions of users provided by the "Multi-user Synchronization" will be needed for harassment prevention and sharing virtual content augmentation among users. Among these building blocks, "Multi-user Synchronization" and "Shared Spatial Mapping and Augmentation" bring challenges that do not exist in single-user scenarios. "Security, Privacy, and Harassment" leads to more research opportunities for the multi-user case, since problems such as harassment prevention are trivial for the single-user case.

A challenging issue when coordinating these components is that some of them share a common objective and may lead to latent conflicts. For example, both "Adaptive Edge Acceleration" and "Remote Content Rendering" aim to address the scalability issue of multi-user XR and will compete for the scarce edge resources, which calls for a judicious design of the scheduling algorithm. Also, there is a natural tension between experience sharing in multi-user XR and security and privacy concerns that will occur during

the interaction of users, creating rich research opportunities for further exploration. Moreover, the encryption of exchanged data among users causes troubles for the offloading of certain operations such as content rendering to the edge that should have no access to the unencrypted data.

1) Programmable mmWave 5G Edge Networking: Delivering immersive content to multiple users requires more network bandwidth than the single-user case [13]. mmWave 5G networks have emerged as a potential solution to enable the high-capacity wireless edge for multi-user volumetric video streaming. However, mmWave comes with the challenge of directional transmission due to high attenuation. When delivering immersive content over mmWave, changes in users' 6DoF motion require continuous adaptation of beams between the access point (AP) and the client (*i.e.*, XR headset). Also, the line-of-sight (LoS) path from the AP to the client can be blocked by users' bodies (*e.g.*, hand or head movement) or other users walking around, which increases the number of beamsearching for maintaining a high signal-to-noise ratio (SNR) and requires additional mechanisms for selecting reflected paths to ensure reliable data delivery. However, the default beamforming in 802.11ad/ay and 5G networks incurs a formidable overhead in dealing with mobility and blockages, resulting in degraded QoE (*e.g.*, impaired visual quality or increased rebuffering time) for multi-user volumetric video streaming.

To this end, we will develop an agile, cross-layer programmable mmWave edge network in CoMIC that can seamlessly integrate user and environmental characteristics specific to immersive content in low-layer wireless control through real-time MAC/PHY reconfigurability. More specifically, we will develop interfaces that can export the information regarding the users (*e.g.*, 6DoF motion predictions) and spatial maps (*e.g.*, presence of ambient reflectors) for fine-grained control of mmWave MAC/PHY. The user behavior and predicted 6DoF motion while consuming immersive content can be leveraged for beam adaptation and switching with low searching overhead. Similarly, the spatial maps can be exploited to determine ambient reflectors that can provide mmWave connectivity in presence of blockages. Moreover, we plan to develop low overhead beam searching strategies by predicting blockages and mobility changes based on interactions among multiple users. These solutions are not only application-ware but also cross-layer because mmWave sensing and parameters such as SNR, beam index, LoS/NLoS, angle of arrival, angle of departure, etc. can improve 6DoF motion prediction in CoMIC, by offering more accurate 6DoF pose estimation of XR devices.

2) Adaptive Edge Acceleration: Immersive technologies such as AR/MR heavily utilize CV algorithms (*e.g.*, object classification and recognition [14]) that are computation intensive and may result in noticeable latency when executed on mobile devices, as they usually benefit from deep learning (DL). Hence, edge offloading is a promising solution for reducing the latency of mobile AR/MR. However, given its large footprint for being close to end-users, the edge is resource-constrained regarding compute and network resources for economic considerations,

The design of SEAR was motivated by our key observation that nearby users may have overlapped views to augment, for example, they may share some common interests when playing together a multi-user AR game.

compared to the centralized cloud infrastructure. This may lead to scalability issues (*e.g.*, prolonged end-to-end latency) when multiple users simultaneously offload their compute-heavy tasks to the edge, a problem that does not exist in the single-user case.

To address the scalability of multi-user AR, we have proposed SEAR [15], a collaborative framework that dynamically splits and schedules AR workload between mobile devices and the edge. The design of SEAR was motivated by our key observation that nearby users may have overlapped views to augment, for example, they may share some common interests when playing together a multi-user AR game. As a result, it is feasible to opportunistically exchange among users the results of offloaded AR tasks and leverage on-device compute resources to relieve the edge workload by designing a collaborative local cache to intelligently reuse these results. Our extensive performance evaluations of SEAR demonstrate that it achieves high object-recognition accuracy for mobile AR and drastically reduces the end-to-end latency compared to the state-of-the-art adaptive edge offloading scheme. We will extend and integrate SEAR into CoMIC by offering generic and adaptive edge acceleration services to multi-user XR. For example, we will design a GPU scheduler to coordinate the CV/DL tasks of different users with the goal of reducing the end-to-end latency and thus improving the QoE of XR.

3) Multi-User Synchronization: To enable the collaboration of users, it is necessary for multi-user XR to accurately synchronize the 6DoF pose of their devices. Otherwise, different users may perceive the same virtual object shared by them at different locations and with different orientations, severely degrading the user experience. For single-user scenarios, popular AR frameworks such as Google ARCore⁴ and Apple ARKit⁵ provide common interfaces to access the 6DoF pose of mobile devices. However, this acquired 6DoF pose cannot be directly applied to multi-user XR, since it is relative to the pose of the device when the AR experience begins.

To overcome this challenge, we will build an edge-assisted approach that enables fast and lightweight multi-user synchronization, which can drastically improve the accuracy while reducing computing and communication overhead, and provide synchronized positions (both relative and absolute) and trajectories. The high-level idea is to leverage multi-user SLAM (Simultaneous Localization and Mapping) by offloading the computation-intensive global map construction to the edge [12]. Our initial work shows that the localization accuracy of multi-user SLAM may not satisfy the requirements of spatial synchronization of the devices' 6DoF pose and the on-device computation overhead is too high to guarantee real-time synchronization. The potential solutions are to resort to the IMU (Inertial Measurement

⁴ <https://developers.google.com/ar> (accessed on 10-August-2022).

⁵ <https://developer.apple.com/augmented-reality> (accessed on 10-August-2022).

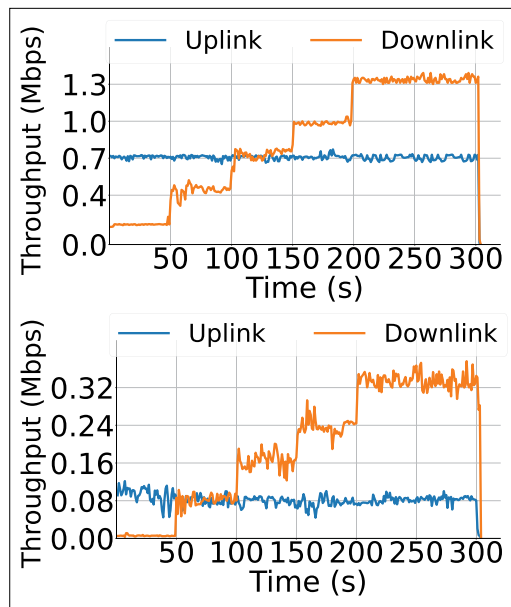


FIGURE 2. Throughput scalability issues of two social VR platforms Horizon Worlds (left) and Mozilla Hubs (right).

Unit) on XR devices to further improve the synchronization accuracy and utilize the pre-built map for XR applications to optimize the on-device operations. Moreover, CoMIC will support real-time content synchronization among users, for example, by sharing manipulation of immersive annotations via the edge.

4) Remote Content Rendering: A key challenge of VR is to render high-quality 3D content at a high frame rate, which is computation intensive. Thus, local rendering on VR headsets requires powerful GPUs that untethered headsets such as Oculus Quest 2 are not equipped with, due to their small form factor. A heavier headset may lead to motion sickness and poor user experience by pulling the user's head forward and down, especially after a long time of use. Modern VR systems switch from rendering all virtual content locally on headsets to offloading the content processing to a server (i.e., remote rendering) [11]. After rendering the content, the server transfers back encoded video frames to VR headsets for display.

Our initial exploration of multi-user VR applications such as popular social VR platforms, including Alt-spaceVR,⁶ Horizon Worlds,⁷ Mozilla Hubs,⁸ Rec Room,⁹ and VRChat,¹⁰ reveals that they still render virtual content such as avatars for embodying users locally. While local rendering works well for the single-user case, it may cause scalability issues when the platform servers need to forward the up-to-date avatar data (e.g., when users move) among multiple users, because the required bandwidth will grow almost linearly with the number of users [4]. For example, Fig. 2 shows how the throughput of Horizon Worlds and Mozilla Hubs changes when more users join a social event at 50, 100, 150, and 200s, respectively. One possible solution to address the scalability issues is to leverage remote rendering, for which the throughput is determined by the visual quality of encoded video frames, instead of the number of users. However, remote rendering

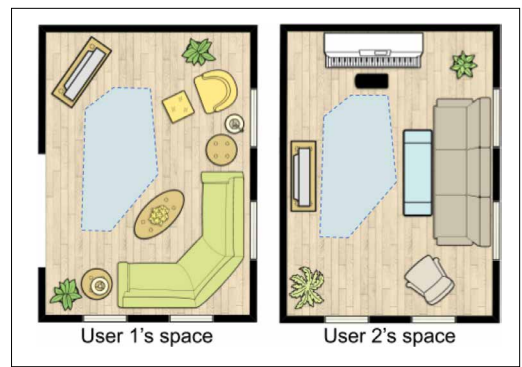


FIGURE 3. Identifying common free space between two users for shared virtual objects.

has its own technical challenges. First, the server should accurately predict users' future viewport as the content rendering and delivery take time. Second, the server needs to render the same number of views as the number of users, since different users may have different views. To address these problems, CoMIC will explore the rich opportunities in multi-user VR and offer an efficient remote content rendering pipeline by jointly predicting the 6DoF motion of multiple users, which may be correlated, and leveraging image-based rendering and neural radiance field (NeRF)¹¹ to synthesize the view of a given user from similar views of nearby users, instead of rendering it from scratch.

5) Shared Spatial Mapping and Augmentation: CoMIC supports multiple users to collaborate simultaneously in a shared environment in AR, where they see and interact with common virtual objects. However, the real, physical environments of users at different sites could be different in terms of the room and furniture layout. Thus, a naive, straightforward approach of visualizing virtual objects in front of users may prompt problems such as occlusion by real-world furniture or a wall.

To overcome such issues, CoMIC will consider the geometry and the semantics of the real-world surroundings of users when augmenting the shared environment with virtual objects. Hence, as shown in Fig. 3, CoMIC will compute a shared space for obstacle-free collaboration among multiple users by considering the free space in the surroundings of each user. A virtual object (e.g., a virtual car) can then be visualized and manipulated in this shared space, which will not induce any occlusion or obstacle collision on any user. Furthermore, suppose the geometries and semantics of the objects in a user's surroundings are obtained through CV techniques (e.g., plane detection and object recognition) on the RGB-D data captured by XR headsets (e.g., Microsoft HoloLens), such information could be leveraged by CoMIC to guide virtual object placement in a contextually-compatible manner, essentially improving the quality of user-perceived experience.

6) Security & Privacy and Harassment & Safety: Similar to any other computing paradigms, immersive computing is subject to various security and privacy attacks and safety concerns. For example, a user's identity could be stolen via hacked VR glasses. There are also new types of challenges in the Metaverse, for example,

⁶ <https://altvr.com> (accessed on 10-August-2022).

⁷ <https://www.oculus.com/horizon-worlds> (accessed on 10-August-2022).

⁸ <https://hubs.mozilla.com> (accessed on 10-August-2022).

⁹ <https://recroom.com> (accessed on 10-August-2022).

¹⁰ <https://hello.vrchat.com> (accessed on 10-August-2022).

¹¹ <https://www.matthew-tancik.com/nerf> (accessed on 10-August-2022).

defending against immersive attacks [2] such as disorientation attacks that may cause collisions with real objects and motion sickness. Even worse, recent news has reported that on social VR platforms, harassment, such as rape, has occurred.

All these accidents and responses suggest that in addition to the emerging demand for security and privacy protections that are similar to the traditional digital worlds, multi-user XR should offer more considerate defense and safety features as one would expect in the real world. To this end, we will build a security, privacy, and safety module in CoMIC that provides privacy-aware mechanisms for multi-user XR, for example, to preserve privacy during the activity tracking of users in social VR. More specifically, we will develop in CoMIC a privacy-preserving continuous and adaptive authentication scheme by employing a federated-learning-based zero-trust framework that authenticates users with multimodal biometric information. We will also explore security and privacy related policies and build a comprehensive harassment detection and prevention framework for sending early alerts, automatically activating personal space, automatically reporting misbehavior, etc.

TESTBED

The CoMIC testbed will be equipped with multiple XR devices such as headsets (e.g., Microsoft HoloLens 2, Magic Leap 2, Oculus Quest 2, HTC VIVE, etc.) and smartphones. To facilitate remote access, we will use the following two types of robots mounted with XR devices for mimicking human behaviors and automating remote experiments. NAO Power V6 Robot is an autonomous, programmable humanoid robot that is widely used to mimic human behaviors, such as walking, sitting down, standing up, etc. NAO can be connected via a wireless connection to update its program. It has a full SDK in Java, C++, Matlab, and Python. LoCoBot is a low-cost, open-source mobile manipulator that runs Facebook AI's PyRobot, a lightweight, high-level interface on top of the robot operating system (ROS). It comes with the capable WidowX 200 Mobile Arm, which offers 5 DoF, high torque, and better durability.

We will equip the testbed with two types of mmWave networks: 60 GHz 802.11ad/ay WiFi and 28 GHz 5G NR cellular networks. Our existing Mason-Milli testbed¹² already includes an extensive number of 60 GHz software radios and 802.11ad COTS devices. We will equip the XR headsets with 802.11ad and new 802.11ay chipsets with multiple phased antenna arrays to support the mmWave communication with the APs. We have already developed MAC/PHY reconfiguration capabilities on COTS 802.11ad devices to change desired beams, extract PHY information, etc. This will enable us to perform full-stack experiments along with MAC/PHY reconfigurability over the mmWave WiFi testbed. In terms of 28 GHz, we will acquire SiversIMA 28 GHz software radio frontend¹³ with USRP X310s¹⁴ and N310s¹⁵ and full-stack 5G NR devices¹⁶ that are compatible with commercial 5G networks but provide reconfigurability through relays. COTS mmWave 5G NR enabled smartphones will be used as XR devices. We will develop cross-layer APIs on software radios using

GNURadio blocks as well as on 5G NR commodity devices using vendor-provided interfaces to the Linux drivers.

TOOLS AND DATASETS

CoMIC will significantly lower the barriers to creating multi-site, multi-user XR applications with a comprehensive suite of tools and datasets.

1. Remote Controller of XR Devices and Robots. CoMIC will allow users to remotely configure and control most XR headsets, depending on the capabilities offered by individual devices. Some XR headsets, for example, Oculus Quest 2, provide tools that enable the test of XR applications by automatically playing back pre-defined inputs.¹⁷ We will extend these tools to facilitate the remote access of XR devices. We will also develop tools for remote XR teleoperation of robots, for example, to control a robot to perform object manipulation tasks remotely.
2. Remote Controller of mmWave 5G Networks. CoMIC will enable users to remotely configure PHY/MAC parameters (e.g., mmWave transmit and receive beams, modulations, etc.) and extract link measurements (e.g., channel state information, SNR, etc.), and provide userspace interfaces.
3. Testbed Customizer. We will build tools that enable remote users to customize the CoMIC infrastructure, for example, by replacing a component in our reference implementation with their own solutions.
4. Collectors of KPIs (Key Performance Indicators). CoMIC will develop tools to collect key metrics of multi-user XR applications, such as communication overhead for synchronization and other QoE metrics for multi-user AR mentioned in the Introduction.
5. Web Portal for User Requests and Service Provisioning. The Web portal will allow community users to submit requests for using the CoMIC sites for remote access or in-person experiments.

Based on CoMIC, we will devise different applications (e.g., XR-based training, healthcare, and data analytics) that feature multi-user XR experiences, from which we can generate novel datasets pertaining to how multiple users interact with each other to perform certain tasks. For example, we can track how multiple users collaborate with each other in a virtual workplace or how they consume volumetric content visualized in XR in a shared physical space. The tracked data comprises 6DoF body pose, 6DoF head pose, eye gaze, etc. Such data would enable us to better analyze and understand how humans collaborate, possibly training generative models that can mimic human collaborations for applications such as synthesizing motions for human-robot collaborations. Furthermore, since CoMIC runs on our edge servers with its agent on XR devices, we will collect both server-side and client-side network traffic traces and resource utilization traces (e.g., CPU, GPU, and memory usage) for multi-user XR.

CASE STUDY

In this section, we present a case study of multi-user volumetric video streaming and demonstrate that it can benefit from CoMIC to improve the QoE.

¹² <http://www.phpathak.com/mason-milli.html> (accessed on 10-August-2022).

¹³ <https://www.sivers-semiconductors.com/sivers-wireless/evaluation-kits/evaluation-kit-evk02001> (accessed on 10-August-2022).

¹⁴ <https://www.ettus.com/all-products/x310-kit> (accessed on 10-August-2022).

¹⁵ <https://www.ettus.com/all-products/usrp-n310> (accessed on 10-August-2022).

¹⁶ <https://airfidenet.com/technology.php> (accessed on 10-August-2022).

¹⁷ <https://developer.oculus.com/documentation/unity/ts-autodriver> (accessed on 10-August-2022).

Although many applications of volumetric videos are specific to multi-user scenarios, limited attention has been paid to multi-user volumetric video streaming, mainly due to its data-intensive nature [7]. In our previous work [13], we find that mmWave WLANs (wireless local area networks) such as 802.11ad can support only a limited number of users for volumetric video streaming and require further optimizations at both the application and PHY/MAC layers to operate as expected. We also find that users share a high degree of viewport similarity when watching volumetric videos, which can be utilized in the multicast transmission to support higher quality videos or scale the system to more users in the immersive viewing session.

In multi-user scenarios, traditional motion/viewport predictions cannot be directly applied as users might occlude the viewport of each other and affect the moving directions. We aim to build a model for joint motion prediction for multiple users to incorporate their interactions. In addition, this multi-user motion prediction could be exploited to estimate the blockage events in mmWave WLANs, which can enable fast, cross-layer, and proactive blockage mitigation. However, building an efficient multicast system to stream volumetric videos is challenging because we need a joint user grouping and scheduling scheme to guarantee an immersive experience for all users. In addition, the directional communication in mmWave WLANs can lead to unbalanced received signal strength (RSS), making the multicast transmission in mmWave WLANs even harder.

To solve these problems, we propose a multicast grouping and scheduling scheme, dubbed M5, to group users with high viewport similarity to improve the multicast performance and conduct one of many possible actions (e.g., prefetching, video quality adaptation, or beam switching) to

offer high QoE for all users. Besides, we employ a new custom beam design to overcome the unbalanced RSS problem. Our preliminary results show that this beam design can effectively improve the multicast transmission performance in mmWave WLANs. Figure 4 (left) shows the RSS improvements that can be achieved by implementing customized multi-lobe beams on CoMIC mmWave radios. We use the programmable mmWave radios equipped with phased antenna arrays of an early prototype of CoMIC to create multi-lobe beams based on the content similarity observed between users at the application layer, which can be calculated using the 6DoF motion prediction. Figure 4 (right) shows the throughput for three scenarios: unicast, multicast with default beams, and multicast with customized multi-lobe beams. We find that multicast with default beams can sometimes perform even worse than unicast due to low common RSS in multicast groups. However, using multicast with customized beams can effectively increase the data rate and further performance gains can be achieved with optimal grouping and scheduling.

We evaluate the performance of M5 and compare it with a baseline scheme that utilizes the default 802.11ad beam-forming to find the best SNR beam for all users and transmit the required volumetric content to them with unicast transmissions and a multi-user extension of ViVo [7] with unicast transmissions through a trace-driven simulation study. We show the frame rate in Fig. 5 (left) and visual quality (in six different levels) in Fig. 5 (right) of the above three schemes with different numbers of users. When the number of users increases, all schemes experience a reduction in frame rate and visual quality because of the frequent blockages among users and the growing size of required content by all users. M5 offers a much higher frame rate than the baseline and ViVo because it can effectively reduce the required bandwidth by leveraging content similarity among multiple users with multicast, instead of sending the same content to individual users via unicast. Furthermore, it can predict the blockage events based on the 6DoF motion prediction of multiple users and prefetch frames to mitigate the blockage events. Due to the same reasons, M5 also outperforms the baseline and ViVo in terms of visual quality.

Note that while frame rate and video quality are not QoE metrics that are specifically designed for multi-user volumetric video streaming, they can clearly demonstrate the benefits of exploring in CoMIC the correlation between multiple users for volumetric video streaming. As an open research problem, there are plenty of opportunities to design QoE metrics and models for multi-user volumetric video streaming and multi-user XR in general, which we are going to investigate in our future work.

DISCUSSION

The development of CoMIC for multi-site, multi-user XR will empower transformative research and offer numerous unique opportunities for different research areas.

Mobile Systems & Computing: The emerging XR applications will consume more bandwidth and limited battery supply on mobile devices,

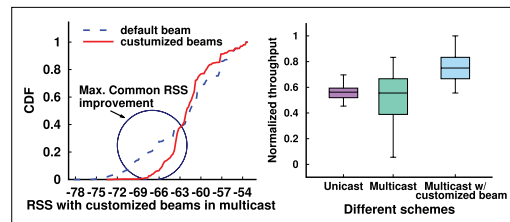


FIGURE 4. (Left) With the customized beam design in multicast, we can efficiently transmit the overlapped cells between users. (Right) Preliminary performance of the multicast system with customized beams.

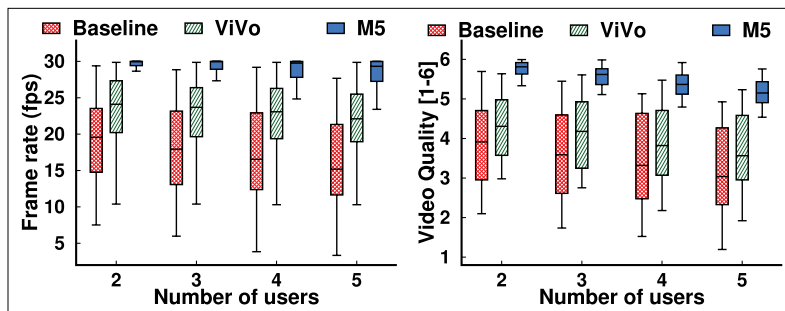


FIGURE 5. M5 improves the frame rate and video quality.

and demand more computation to provide a truly immersive experience. To overcome these challenges, CoMIC will enable tremendous new research opportunities, for example, multi-user 6DoF motion prediction, collaborative content encoding/decoding, cooperative local and remote rendering, scalable volumetric video streaming, etc.

Wireless Networking: While the next generation of wireless networks operating at mmWave frequencies can provide multi-gigabit link capacities, these networks are likely to be strained in multi-user scenarios where each user expects a high-quality immersive experience. This combined with the directional nature of mmWave links, which requires continuous adaptation in presence of mobility and blockages, creates an unprecedented landscape of wireless networking related challenges. CoMIC will provide a first-of-its-kind framework that will enable wireless researchers to better understand and evaluate these interactions and solve the fundamental wireless networking challenges for collaborative mobile immersive computing.

Security & Privacy: Although harassment and safety have not been a central focus yet, with multi-site, multi-user XR applications, such problems are quickly emerging and demand effective solutions, indicating security, privacy, and safety cannot be treated as an afterthought in the system design. Solutions will be needed for privacy-preserving digital forensics, automatic crime detection, reporting, and punishment, harassment prevention and moderation, digital assets tracing, etc. CoMIC provides an effective channel to test out these solutions.

CoMIC will appeal to the other research areas, including computer graphics, computer vision, human-computer interaction, and robotics, by stimulating new research directions on distributed rendering, collaborative content creation, interactive 3D modeling, human pose tracking, scene understanding, human-robot interactions, etc.

CONCLUSION

In this paper, we propose CoMIC, a first-of-its-kind infrastructure for conducting multi-user XR research. CoMIC consists of a software framework, a physical testbed, and a set of tools and datasets for serving the research community. It features a collaborative, visual-first, and hologram-based computing space and a QoE-driven, multi-site, and immersive communication framework. Through a case study on its early prototype, we demonstrate that CoMIC can effectively scale multi-user volumetric video streaming over mmWave by increasing the number of supported users. We hope the full-fledged CoMIC can facilitate and advance the future research of collaborative mobile immersive computing.

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